[Prop. Let $V=\{f:[-1,1]\to\mathbb{R} \mid f \text{ is continuous}\}$ with equipped an inner product

$$\langle f,g\rangle=\int_{-1}^{1} f(t)\cdot g(t)\,dt$$

Let W be the subspace of odd functions. Find $W^{\perp}$.

Prof. $f\in W^{\perp} \iff$ for any odd function g, $\int_{-1}^{1} f(t)g(t)\,dt=0$

Since for any even function g and any odd f, fg is odd, so $\int_{-1}^{1} fg=0$.

That is, $W^{\perp}$ contains all even function. Moreover, for any $f\in V$,

$$f(x)=\underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even}}+\underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd}}$$

So, for subset W' of V consisting by all even functions,

$$V=W+W', \text{ and } W'\subseteq W^{\perp}.$$

Now, clearly $W\cap W'=\{0\}$, $V=W\oplus W'$.

If $W'\neq W^{\perp}$, then, for some $h\in W^{\perp}$, $h\notin W'$. Again, h can be decomposed as

$$h(x)=\underbrace{\frac{h(x)+h(-x)}{2}}_{\in W'}+\underbrace{\frac{h(x)-h(-x)}{2}}_{\in W}$$

Since $W'\subset W^{\perp}$, $h(x)-\frac{h(x)+h(-x)}{2}\in W^{\perp}$, and $\in W$.

But, this implies that it is contained in $W\cap W^{\perp}$, so it is zero:

If $x\in W\cap W^{\perp}$, then $\langle x,x\rangle=0$, or $x=0$.

Thus, $h(x)-\frac{h(x)+h(-x)}{2}=\frac{h(x)-h(-x)}{2}=0$, so $h(x)=h(-x)$, even. Contradicts to $h\notin W'$.

Now, we can construct a projection of V onto W as:

$$P:V\to V: f(x)\mapsto \frac{f(x)-f(-x)}{2}$$